Chapter 5: Decision Making & Branching in Java

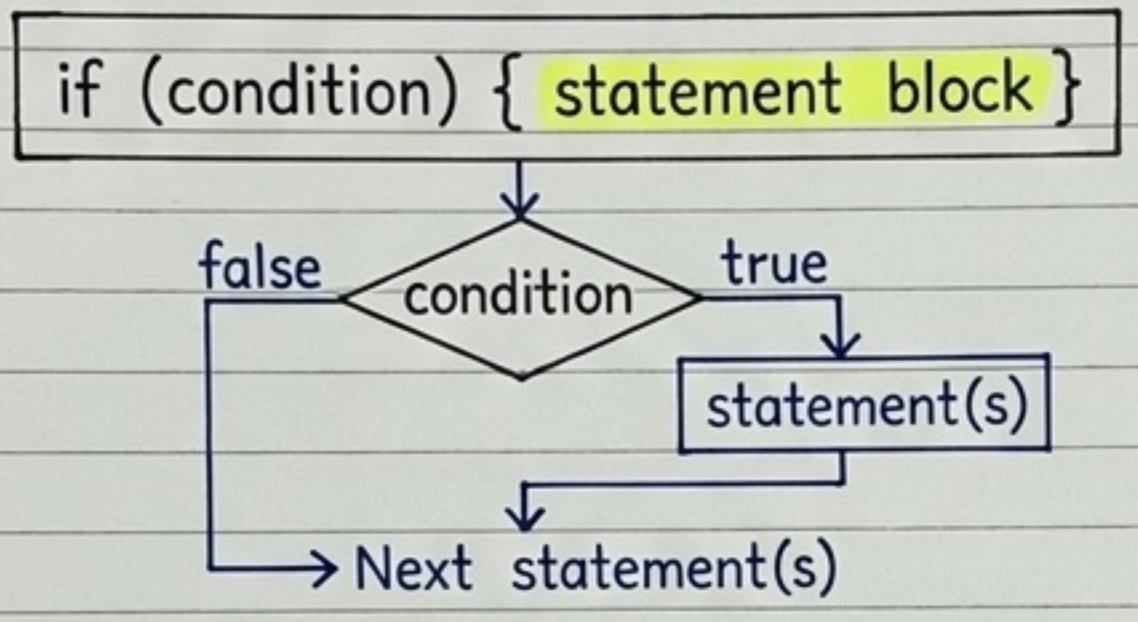
1. Types of Decision Making Statements in Java

- if statements
- switch statement
- conditional operator (?:)

2. Types of IF Statements

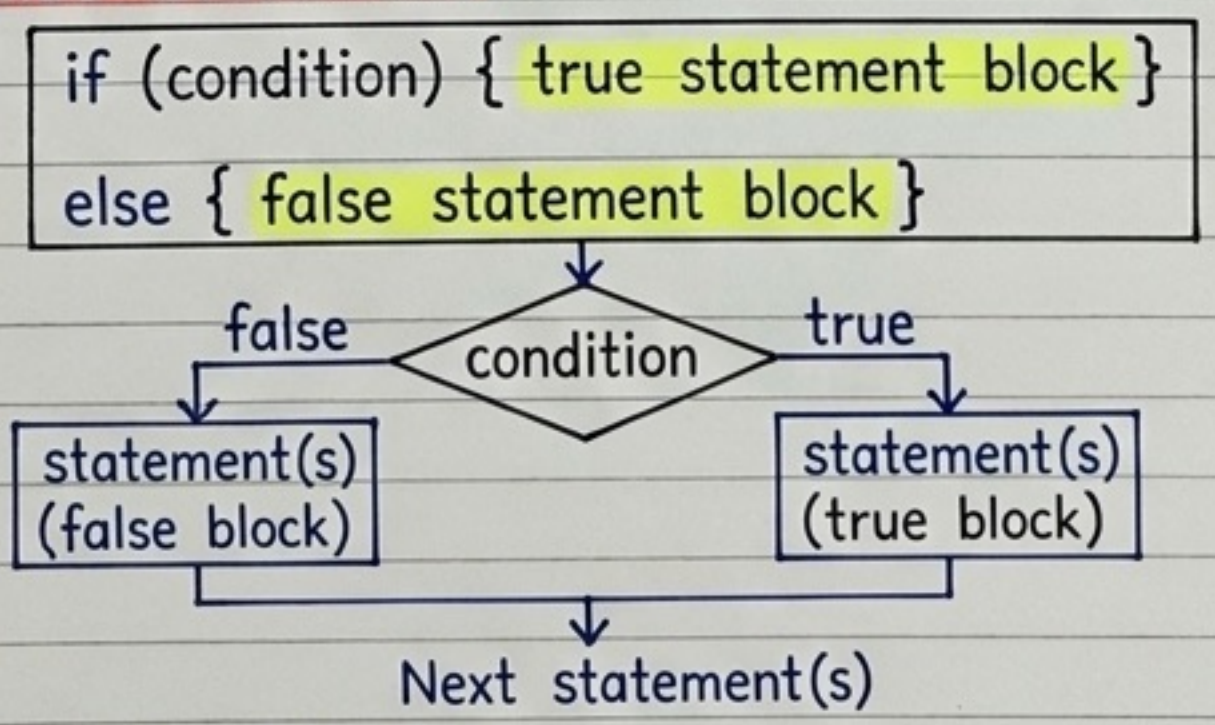
- Simple if
- If..else
- Nested if
- If..else if ladder

3. Simple IF Statement




```
Example: int a=20, b=10;
         if(a>b) System.out.println("A is Largest");
       // Output: A is Largest Number
```

4. If..Else Statement




```
Example: int a=20, b=10;
         if(a>b) System.out.println("A is Largest");
         else System.out.println("B is Largest");
       // Output: A is Largest Number
```

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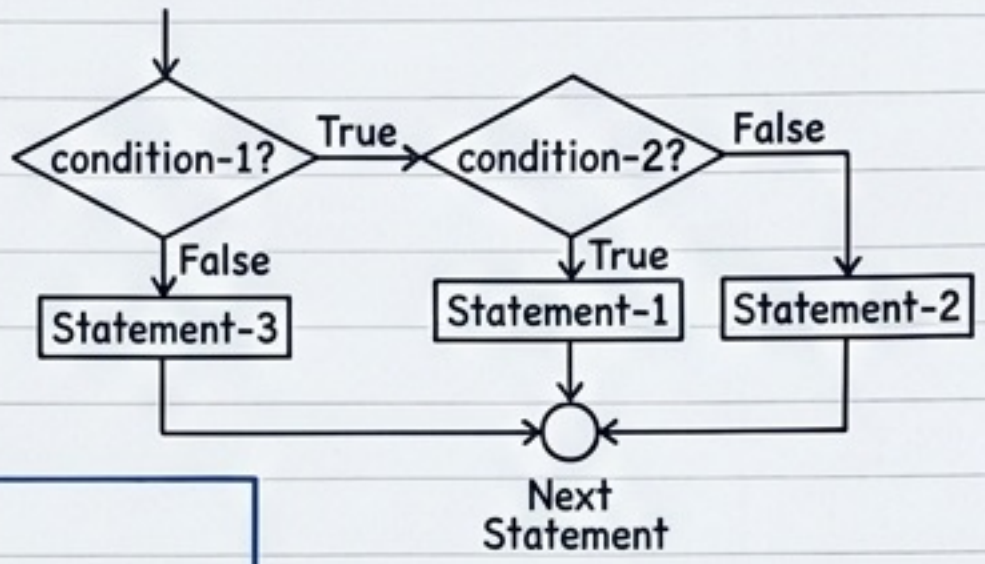
Chapter 5: Nested-If and Else-If Ladder in Java

5.2.3 The Nested-If Statement

General Syntax:

```
if (condition-1) {
      if (condition-2) { statement-block-1; }
      else { statement-block-2; }
} else { statement-block-3; }
```

Inner If



Code Example Snippet:

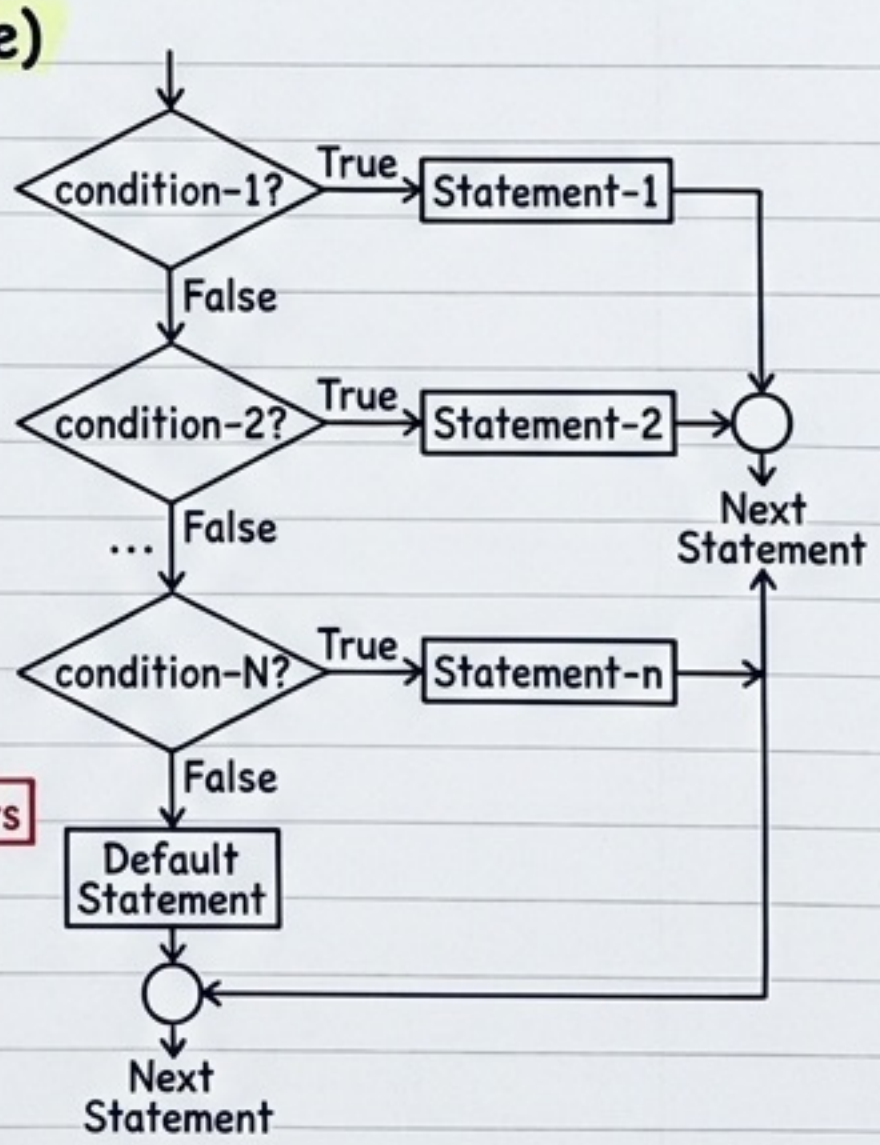
```
// Largest of 3 Numbers
class largest {
    public static void main(String[] args) {
        int a=20, b=10, c=30;
        if(a>b) {
            if(a>c) {
                System.out.println("A is Largest Number");
            } else {
                System.out.println("C is Largest Number");
            }
        } else {
            if(b>c) {
                System.out.println("B is Largest Number");
            } else {
                System.out.println("C is Largest Number");
        }   }
    }
}
```

→ Output: C is Largest Number

5.2.4 The Else-If Ladder (if..else if..else)

General Syntax:

```
if (condition-1) { statement-block-1; }
else if (condition-2) { statement-block-2; }
...
else if (condition-N) { statement-block-n; }
else { default-statement-block; }
```



Code Example Snippet:

```
// Grade of Student
double average=85;
if (average>=80) {
    System.out.println("Honours");
} else if (average>=60) {
    System.out.println("First Division");
} else if (average>=50) {
    System.out.println("Second Division");
} else {
    System.out.println("Fail");
}
```

→ Output: Honours

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Chapter 5: The Switch Statement & Comparison in Java

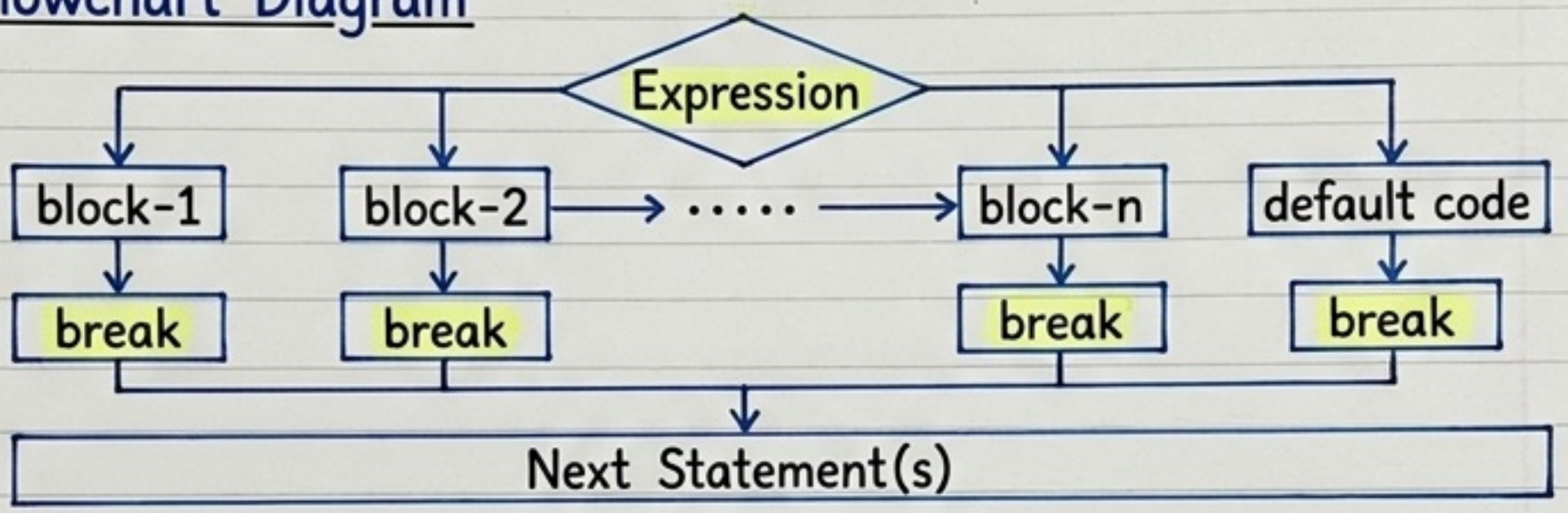
5.3 The Switch Statement in Java

- **Syntax & Rules**

```
switch (expression) {
    case value-1: block-1; break;
    case value-2: block-2; break;
    default: default-block;
}
```

Rules: - switch variable must be integer, char, or String;
- case values must be constants;
- break statement terminates switch execution;
- default case catches non-matching values.

Flowchart Diagram



Programming Example - Menu-Driven Calculator

```
int ch, a=10, b=2;
Scanner sc = new Scanner(System.in);
System.out.println("Enter choice: 1(+), 2(-), 3(*), 4(/)");
ch = sc.nextInt();
switch(ch) {
    case 1: System.out.println("Sum: " + (a+b)); break;
    case 2: System.out.println("Prod: " + (a-b)); break;
    case 4: System.out.println("Quot: " + (a/b)); break;
    default: System.out.println("Invalid choice");
}
```

5.4 Difference between if and Switch Statement

Expression Types:	if (any boolean type)	switch (int, char, String)
Decision Structure:	if (binary true/false)	switch (multi-way choices)
Break Usage:	if (no break by default)	switch (requires break to prevent fall-through)
Best Use Case:	if (top-down logic)	switch (menu-driven programming)
Else/Default position:	if (else at the end)	switch (default anywhere)

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